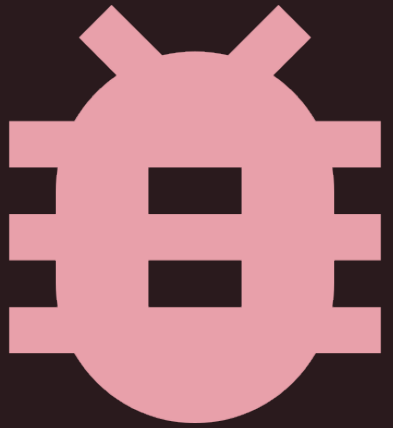
LAB - DAY 6

Higher National Diploma in Network Engineering, Year 2
BSc (Hons) Ethical Hacking & Network Security, Year 2

Find and fix injection and scripting flaws

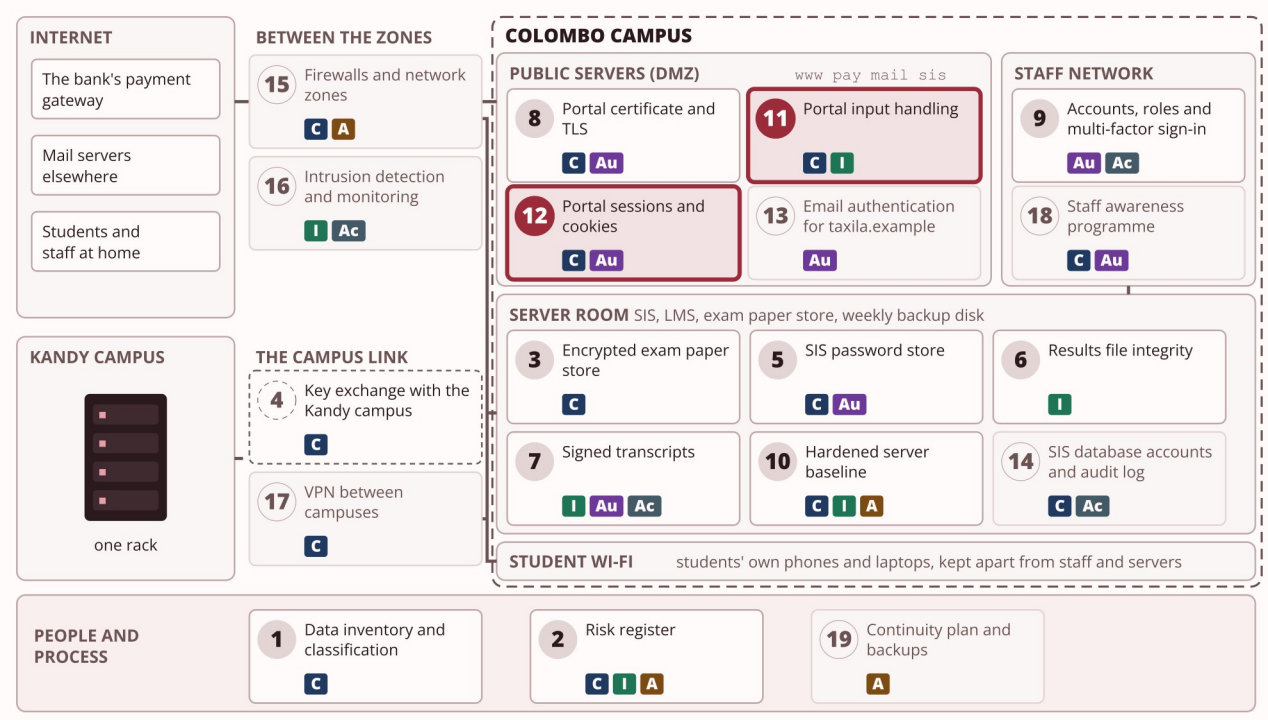


Adds to the register: **11** **12**

Floor tier: every step | **Full tier: recommended**

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The security register



TODAY

11 Portal input handling

12 Portal sessions and cookies

KEY

- Added today
- Added earlier
- Concept only
- Not yet added

IT

HEAD OF IT

Find me the line that trusts what a student types, and show me it fixed.

The Head of IT, at the start of the lab.

Before you start

```
# both tiers: your index number seeds your own practice record
# floor tier: open the sqli-login and output-encoding demos
# full tier: snapshot day-6-start, then download taxila-portal.py
```

You never craft an attack and you never type one. A prepared test probes; you read the report, find the line, and fix it.

SEGMENT 1 - 40 MIN

Read the login code and find the injection flaw

Both tiers

Run the test and find the injection flaw

1 Run the provided test with your index; read its report.

Expected: The probe returns 6 of 6 records; a normal sign-in returns 1.

Floor path: Run the login test in `sqli-login`.

2 Open the code; find the line that joins input into the query.

Expected: The marked flawed line inside `flawed_login`.

Floor path: Step the test value through the joined query in the demo.

```
$ python3 taxila-portal.py --test --index COHNDNE999X000
the prepared login probe returned 6 of 6 records
RESULT: OPEN. The login is injectable.
```

The test value is prepared and never shown as a ready string. The report shows only the record counts.

SEGMENT 2 - 45 MIN

Fix the query and prove it with the test

Both tiers

Apply the parameterised query and prove it

3 Set `SAFE_LOGIN = True`; save; run the test again.

Expected: The probe returns 0 records; a normal sign-in still returns 1.

Floor path: Choose the parameterised query in the demo and run the test.

```
$ python3 taxila-portal.py --test --index COHNDNE999X000
the prepared login probe returned 0 of 6 records
RESULT: CLOSED. A normal sign-in still works.
```

SEGMENT 3 - 40 MIN

Find and fix the echo flaw

Both tiers

Find and fix the echo flaw

4 Run the note test; find the echo line; set `SAFE_ECHO = True`.

Expected: Before: the bold note reaches the page as markup. After: shown as plain text, with your index.

Floor path: Toggle encoding in `output-encoding`.

```
Note page: the bold note reached the page as markup (OPEN)
after output encoding: shown as plain text (CLOSED)
```

The note is bold text only. A script in a note would run in the same place; encoding shows it as text instead.

SEGMENT 4 - 25 MIN

Prove it, break it, reset and write up

Both tiers

Prove it, then break it

PROVE IT

- Login test: 6 records before the fix, including your seeded record
- Login test: 0 records after, 1 for a normal sign-in
- Note test: page changed before encoding, plain text after

BREAK IT

- Set `SAFE_LOGIN` back to `False`; run the test
- The probe returns all 6 records again
- Set it to `True`; the leak is closed

Reset, and the rules

RESET

- Keep your two test reports and your register lines first
- Full tier: restore `day-6-start`
- Floor tier: reload the two demos

RULES OF ENGAGEMENT

- No tool in this lab can affect anyone else's system
- A prepared test probes; you read, find and fix

Record your evidence

EVIDENCE

- Your seeded record and both login counts (6 before, 0 after)
- The test report before and after your query fix, showing your seeded record leaking and then not
- The flawed line and your fixed line
- The echo test before and after output encoding
- The hand-drawn code-and-data sketch
- Your 120-word explanation

Questions & Discussion

Next: Day 7, email and databases. You see why an email cannot prove who sent it.



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